

SYSTEM FLOWCHARTS

ALUMNI WEB APPLICATION

Process Flows & System Architecture

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01. Introduction

This document provides comprehensive flowcharts and process diagrams for the Alumni Web Application. Each flowchart illustrates the logical flow of operations, decision points, and data movement throughout the system.

1.1 Purpose

These flowcharts serve multiple purposes including system documentation, developer reference, training materials for administrators, and quality assurance testing guidelines. They provide a visual representation of how users interact with the system and how data flows through various components.

1.2 Flowchart Symbols Legend

Symbol	Name	Description
()	Start/End	Represents the beginning or end of a process
[]	Process	Represents a process, action, or operation
{ }	Decision	Represents a decision point with Yes/No or multiple options
→	Flow Direction	Indicates the direction of process flow
(D)	Database	Represents database operations or storage

02. Main System Flowchart

The main system flowchart illustrates the complete user journey through the Alumni Web Application, showing all major entry points, user roles, and their respective workflows.

2.1 Entry Points

START	Alumni Web Application Launch
↓	
Entry Point?	Three possible entry points: 1. Public User → Home Page 2. Admin → Admin Login 3. Alumni Member → User Login

2.2 Public User Flow

- User lands on Home Page
- Views featured alumni profiles in carousel
- Sees alumni network statistics
- Clicks Register button
- Fills registration form with personal details
- Uploads photo (max 5MB, JPG/PNG/GIF)
- Creates username and email
- Sets secure password
- Submits registration
- Status set to 'Pending Admin Approval'
- Waits for admin review and approval

2.3 Admin User Flow

2.3.1 Login Process

Step 1	Admin navigates to Admin Login page
Step 2	Enters admin credentials (username/email and password)
Decision	Valid credentials? • If NO → Show error message → Return to login • If YES → Generate JWT token → Redirect to Admin Dashboard

2.3.2 Admin Dashboard Actions

Once logged in, admin can choose from the following actions:

Action	Process	Outcome
View Statistics	Display dashboard with counts	Shows total, employed, self-employed, studying
Add Alumni	Open form, enter details, upload photo	New alumni record saved to database

Action	Process	Outcome
Manage Alumni	View list, search, edit or delete	Alumni records updated or removed
Manage Users	Create login, reset password	User accounts created or modified
Approve Registrations	Review pending, approve/reject	Account created or registration rejected

2.4 Alumni User Flow

2.4.1 Login Process

- Alumni member clicks User Login
- Enters credentials (username/email and password)
- System validates credentials
- If invalid → Show error → Return to login
- If valid → Generate JWT token → Redirect to User Dashboard

2.4.2 User Dashboard Actions

- **View Alumni Directory:**

Browse all approved alumni profiles with view-only access

- **Search Alumni:**

Use advanced search with filters for name, profession, company, batch, department, or location

- **My Profile:**

View personal profile information and change password

- **Contact Alumni:**

Send messages or emails to connect with fellow alumni

03. Authentication & Security Flowchart

This flowchart details the complete authentication process including security measures, token management, and role-based access control.

3.1 Authentication Process Overview

Step	Process	Details
1	User Visits Site	System checks for existing authentication token
2	Authentication Check	If authenticated → Check role → Route to dashboard If not authenticated → Show login options
3	Enter Credentials	User enters username/email and password in login form
4	Rate Limit Check	System checks for excessive login attempts If too many attempts → Block request (429 error) → Wait cooldown If OK → Proceed to validation
5	Validate Credentials	Query database for username/email If user not found → Show error → Return to login If user found → Proceed to password verification
6	Password Verification	Use bcrypt to compare entered password with stored hash If password incorrect → Show error → Return to login If password correct → Check account status
7	Account Status Check	Verify account is active (is_active = true) If inactive → Show inactive message → Return to login If active → Generate JWT token
8	Generate JWT Token	Create JWT token with user ID, role, and expiration time Token payload: {userId, role, exp: 24h} Sign token with secret key
9	Set Session Cookie	Store JWT token in HTTP-only cookie Set secure flag for HTTPS transmission Update last_login timestamp in database
10	Redirect to Dashboard	Check user role from token If role = admin → Redirect to Admin Dashboard If role = user → Redirect to User Dashboard

3.2 Security Measures

Security Feature	Implementation
Password Hashing	bcrypt algorithm with 10+ salt rounds Passwords never stored in plain text One-way hashing prevents password recovery
JWT Token Authentication	Stateless authentication mechanism Token expires after 24 hours Signed with secret key to prevent tampering Includes user ID and role for authorization
Rate Limiting	Maximum 5 failed login attempts per 15 minutes Prevents brute force attacks Temporary account lockout after threshold
HTTP-Only Cookies	Tokens stored in HTTP-only cookies Not accessible via JavaScript (XSS protection) Secure flag ensures HTTPS-only transmission
HTTPS/TLS Encryption	All data transmitted over encrypted HTTPS SSL/TLS certificate required Protects credentials during transmission
Role-Based Access Control	Admin role: Full CRUD access to all features User role: View-only access to alumni directory Middleware validates role on every request

3.3 Session Management Flow

- User makes authenticated request
- Middleware extracts JWT token from cookie
- Verify token signature and expiration
- If token expired → Check refresh token availability
- If refresh available → Issue new token → Continue session
- If no refresh → Force logout → Redirect to login
- If token valid → Extract user data → Allow request
- Check role permissions for requested action
- If authorized → Process request → Return response
- If unauthorized → Return 403 Forbidden error

3.4 Registration and Approval Flow

Step 1	User fills registration form with personal and professional details
Step 2	Uploads photo (validated for size max 5MB and type JPG/PNG/GIF)
Step 3	Creates password (bcrypt hashes password with 10 salt rounds)
Step 4	System saves registration with status 'Pending Approval' to database
Step 5	Admin reviews registration in approval queue
Decision	Admin Decision: If APPROVED:

	• Create user account in users table • Link user account to alumni record (alumni_id foreign key) • Send email with login credentials • User can now login with credentials If REJECTED: • Send rejection notification email • User can re-register if desired
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3.5 Logout Process

4. User clicks logout button
5. System clears JWT token from memory
6. Delete session cookie
7. Clear any cached user data
8. Redirect user to home page
9. User must login again to access authenticated features

04. System Architecture & Data Flow

This section describes the three-tier architecture of the Alumni Web Application, showing how data flows between the client, server, and database layers.

4.1 Three-Tier Architecture Overview

Layer	Technology	Responsibilities
Client Layer (Presentation)	React with Vite Material-UI / Tailwind CSS React Router	- User interface rendering - Client-side routing - Form validation - State management
API Layer (Business Logic)	Node.js Express.js framework JWT authentication Multer (file uploads)	- RESTful API endpoints - Business logic processing - Authentication & authorization - Data validation
Database Layer (Data Storage)	MySQL 8.0 Sequelize ORM Connection pooling	- Data persistence - Query execution - Transaction management - Data integrity

4.2 Request-Response Data Flow

Example: Adding a New Alumni Record

10. CLIENT:

Admin fills out 'Add Alumni' form in React interface
 Uploads alumni photo
 Client validates form data (required fields, email format, etc.)
 Sends POST request to /api/alumni with form data and photo

11. API LAYER:

Express Router receives POST request
 Rate Limiter checks request frequency
 Auth Middleware verifies JWT token and admin role
 Validation Middleware sanitizes and validates input data
 Multer processes photo upload and saves to storage
 Alumni Controller receives validated data

12. DATABASE LAYER:

Controller executes INSERT query on alumni table
 MySQL validates data against schema constraints
 Record saved with auto-generated ID
 Database returns success response with new record ID

13. RESPONSE:

Controller formats success response
 Express sends JSON response to client
 Client receives response and shows success message
 Dashboard statistics updated automatically
 UI redirects to alumni list showing new record

05. Admin CRUD Operations Flowchart

This flowchart details all Create, Read, Update, and Delete operations available to administrators for managing the alumni database.

5.1 CREATE Operation

Process Flow for Adding New Alumni:

Start	Admin clicks 'Add New Alumni' button from dashboard
Step 1	System displays Add Alumni form with empty fields
Step 2	Admin fills required fields: • Name (required) • Email (required, unique) • Phone number • Profession • Current company/organization • College name • Department • Batch year (graduation year) • Current location • Employment status (Studying/Employed/Self-Employed)
Step 3	Admin uploads alumni photo
Validation	Photo Validation: • Check file size (max 5MB) • Check file type (JPG, PNG, GIF only) • If invalid → Show error message → Return to upload step • If valid → Proceed to form validation
Validation	Form Validation: • Check all required fields are filled • Validate email format and uniqueness • Validate phone number format • If validation fails → Show specific error messages → Return to form • If validation passes → Proceed to save
Step 4	System saves photo to file storage (local filesystem or AWS S3)
Step 5	Execute database INSERT query: INSERT INTO alumni (name, email, phone, profession, company, college, department, batch_year, location, status, photo, created_at) VALUES (?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, NOW())
Success	Display success message: 'Alumni record created successfully' Update dashboard statistics (total count, status count) Redirect to alumni list or dashboard

5.2 READ Operations

5.2.1 View All Alumni

- Admin clicks 'Manage Alumni' from dashboard
- System executes: `SELECT * FROM alumni ORDER BY created_at DESC`
- Display alumni list in table format with key columns
- Show action buttons (View, Edit, Delete) for each record
- Display pagination controls if records exceed page limit

5.2.2 View Single Alumni Details

- Admin clicks 'View' button on alumni record
- System executes: `SELECT * FROM alumni WHERE id = ?`
- Display complete profile with all fields
- Show alumni photo
- Provide options to Edit or Delete from detail view

5.2.3 Search and Filter

- Admin opens search interface
- Selects search criteria (name, profession, company, batch, department, status, location)
- System builds dynamic SQL query with WHERE clauses
- Executes query: `SELECT * FROM alumni WHERE [conditions] ORDER BY name ASC`
- Display filtered results in table
- If no results → Show 'No alumni found' message
- Provide export to CSV/Excel option

5.3 UPDATE Operation

Start	Admin selects alumni to edit from list or detail view
Step 1	Fetch current alumni data: <code>SELECT * FROM alumni WHERE id = ?</code>
Step 2	Populate edit form with existing data
Step 3	Admin modifies desired fields
Decision	Change photo? If YES: - Admin uploads new photo - Validate new photo (size, type) - If invalid → Show error → Return to upload - If valid → Delete old photo from storage - Save new photo to storage - Update photo path in form data If NO: - Keep existing photo path - Proceed to validation
Validation	Validate all modified fields If validation fails → Show errors → Return to form If validation passes → Proceed to update
Step 4	Execute UPDATE query:

	UPDATE alumni SET name=?, email=?, phone=?, profession=?, company=?, college=?, department=?, batch_year=?, location=?, status=?, photo=?, updated_at=NOW() WHERE id=?
Success	Display success message: 'Alumni record updated successfully' Update dashboard statistics if status changed Redirect to alumni list or detail view

5.4 DELETE Operation

Start	Admin selects alumni to delete and clicks 'Delete' button
Step 1	System displays confirmation dialog: 'Are you sure you want to delete this alumni record? This action cannot be undone.'
Decision	Confirm Delete? If NO (Cancel): • Close dialog • Return to alumni list • No action taken If YES (Confirm): • Proceed to deletion process
Step 2	Check for related data: SELECT * FROM users WHERE alumni_id = ?
Check	Has User Account? If YES: • Delete user account first (cascade delete) • DELETE FROM users WHERE alumni_id = ? If NO: • Proceed directly to alumni deletion
Step 3	Delete photo file from storage: Get photo path from alumni record Delete file from filesystem or S3 bucket
Step 4	Delete alumni record from database: DELETE FROM alumni WHERE id = ?
Success	Display success message: 'Alumni record deleted successfully' Update dashboard statistics (decrease count) Refresh alumni list (remove deleted record) Redirect to alumni list

06. Conclusion

These flowcharts provide a comprehensive visual representation of the Alumni Web Application's functionality and data flow. They serve as essential documentation for development, testing, training, and maintenance purposes.

6.1 Key Takeaways

- **Three-Tier Architecture:**

Clear separation between client, API, and database layers ensures maintainability and scalability

- **Robust Security:**

Multiple security layers including bcrypt password hashing, JWT authentication, rate limiting, and HTTPS encryption

- **Role-Based Access:**

Clear distinction between admin full-access and user view-only permissions

- **Complete CRUD Operations:**

Administrators have full control over alumni data with proper validation and error handling

- **Data Integrity:**

Comprehensive validation at multiple levels ensures clean and accurate data

6.2 Usage Guidelines

For Developers:

- Use these flowcharts as reference during development
- Ensure all decision points and validations are properly implemented
- Follow the data flow patterns for consistent implementation

For Testers:

- Create test cases based on each flow path
- Verify all decision points handle both positive and negative scenarios
- Test error handling and validation messages

For Administrators:

- Understand the complete workflow of the system
- Reference these flowcharts during training sessions
- Use as troubleshooting guide when issues arise

These flowcharts will be updated as the system evolves to reflect new features and modifications.

Contact Information

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